

EXP2-1 – The same or different

Comparing data and ensuring that data is comparable!

The **aim** of this task is to develop tools and skills, in order to:

- Present and report data in the correct way (units, significant figures, error, and others)
- Determine whether different samples have the same characteristics or properties, whether they behave differently, and whether these differences are statistically significant

The **objectives** of this exercise are:

- Check, analyze and correct the data gathered during the Experiment in “Introduction in Engineering”
- Consider the errors of the data and present a short report
- Critically compare and discuss the results of different samples
- Present a statistics analysis report

For this task you will use the data that previous students gathered during Experiment in ‘Introduction in Engineering’. In particular, elongation by weight of your unmodified Sylgard 184, and data produced from samples using Sylgard 184 modified with Urefil-3 (a type of additive), both from the Materials and Polymers Programmes. Please note that numbering system for the data in these files does not correspond in any way to the groups that generated it.

The data in the files is the raw data, as it was presented by students. Since the focus of this Experiment is to explore tools to compare data, for the sake of time, the Young’s moduli of the samples have already been calculated for you. **However, it is your responsibility to check that these calculations are correct.** In addition, not all the data may have been collected or presented in the correct format. Therefore, it is also your responsibility to check if there are issues with the data, and decide what to do, if this is the case. Please let your teachers know if there are any data that needs adjusting or removing from the data set, and be prepared to justify your decision.

The following concepts will be relevant for this task:

- mean
- median
- mode
- normal distribution
- standard deviation
- standard error
- regression
- coefficient of determination (R^2)
- t-test
- z-test
- significant figures
- statistically significant
- frequency
- histogram
- binning and bins (in the context of histograms, not in the context of rubbish!)

The **learning outcomes** for this task are to understand, and be able to work with the concepts above. In particular how to calculate the relevant statistics shown above using Excel, and how to compare different samples using different statistical methods. Professional skills associated with this task are time and group management, independent study and research, and report writing.

What you should do:

Part 1 Represent the data in a correct way

Group management meeting 1 (pre-session tasks):

Working with your groups:

1. Look at the instructions for the Experiment in Introduction in Engineering Materials (uploaded on the QMPlus page for this module), recall the procedure of the experiment and identify what major sources of error may appear in the experiment. Make a note of these to discuss in class.
2. Look at data. There are three files provided (Excel Spreadsheets), these are selected data for:
 - unmodified Sylgard 184 samples from the Materials Programme
 - unmodified Sylgard 184 samples from the Polymer Programme
 - Urefil-3-modified samples from both Materials and Polymer programmes.

Write down comments about the quality of the data presented, if relevant. There is likely to be data that may present issues for analysis, as not all groups were equally careful about how they gathered, recorded and presented the data. Discuss possible choices about what to do with such data, if anything, and be prepared to justify your choices.

Important items to discuss: **search for outliers, search for ways the data was presented that is incorrect and may affect the results.** But always remember, **data is data, and errors are not mistakes.** The data presented were taken as they were provided by student groups to teachers, without further modification: how good were all groups in making sure the data were recorded/entered/typed correctly and usable to you?

3. Read carefully the instructions for the GMM2 & GMM3 tasks below. Then, prepare a new Excel spreadsheet with the modified data after your justification that you will need to analyse during the task. This Excel spreadsheet will be submitted together with your report at the end of the task as supplementary, including new sets of data, all calculations processes (functions should be kept) and graphs that you have carried out.
4. Apply what you have learned in PDP classes, make a suitable time management and group management plan to carry out the tasks within the time allocated. Part of the assessment for this task will be how well you have collectively managed the time, the task and the group work. Make sure you make your own effort to contribute to the group work during supervision. Start writing your report, in your group, as you go along!

Group management meeting 2 (please be prepared, you may initiate work ahead of the day):

Working in your groups, consider the following discussion/questions, for which you will need to make calculations using Excel spreadsheets and start writing a report.

Section 1: A discussion of the data as presented (see GMM1), including identification of data that presented serious **issues/mistakes** (not errors), and what was your decision about it, including a short justification.

(No matter if you decide to delete, change or keep the data, you need to show your justification/evidence.)

Section 2: What parameter(s) give you confidence about how representative of the actual samples (real) the calculated modulus is? (It would be better if include examples as comparison)

Section 3: What's the **error** in the calculated modulus? The data were expressed as calculated by Excel without any further modifications: please calculate the **mean, mode, median, SD, SE and CI** for all the samples separately, and express the values with **significant figures**. What are the units of the calculation in the original table? What units would be best to use and why? How relevant is the original measurements (raw data without any modification) being correct compare with the data after your modification/correction in each calculated modulus?

Section 4: What can you say about the data samples (Materials and Polymers): do these samples follow a normal distribution? How can you investigate this in a graphic manner?

Part 2 Significance tests**Group management meeting 3 (please be prepared, you may initiate work ahead of the day):**

Working in your groups, consider the following discussion/questions, for which you will need to make calculations using Excel spreadsheets and complete writing the report:

Section 5: Are the data from the unmodified Materials and Polymer samples comparable?

Section 6: Are the data from the Urefil-3-modified Materials and Polymer samples comparable?

Section 7: Are the data between unmodified and modified samples comparable?

For the 3 sections above: If they are comparable, at least two methods should be employed, one of which should be by the means of graphics; if they are not comparable, please analyse and explain why, and you are not expected to use other more complex statistics analysis tools/methods out of the session at this stage.

Section 8: A brief discussion of how you dealt with problem solving and group management (correlate with PDP).

Group management meeting 4 (post-session tasks):

Finish the report containing a selection of essential graphs that support your discussion in GMM3, further graphs can be appended on the Excel spreadsheet. (*Your report must be a complete report even without Excel spreadsheet as supplementary) You will submit this report, together with only **1 Excel file** that contains all your relevant calculations and graphs. (Do not re-submit the original Excel spreadsheets that you have been provided on QMPlus, in total, or partially, just an Excel spreadsheet that shows your work in this task).

The report will be **no more than 1800 words** (exclude references) containing **8 sections** you discussed in **GMM2 & GMM3** tasks. You can give subtitles of each section as you wish, but make sure the structure of your report follows a reasonable logic order. You may use narrow page margins, letters no smaller than font size 11 and line spacing no less than 1.15.

The **title** of the report will be: **EXP2-1 Same or different Report XX** (where XX is your group number, e.g. M16 or P8). The names of **all the group members** should be given as authors of the report.

You will **submit** your group's report as a **pdf document on QMPlus** using the file name:

'QXU5017 EXP2-1 Report XX' (where XX is your group number, e.g. M9 or P17).

The report must be completed and **submitted by 17:00 on Sunday 5th March 2023.**

The report must be the **work of the whole group** and the content agreed by all group members.

All group members must hit the submit button to submit the report.

Marks breakdown:

EXP2-1 is 20% of the marks for QXU5017

Marks will be deducted for late submissions, or submissions that have not followed instructions precisely.

Schedule:

M Groups

Date:	Time:	Place:	Activity:
Monday 20 th	10:30-11:15	QMES Floor 1	Introduction to QXU5017 & EXP2-1 Part1
Tuesday 21 st	14:00-15:40		Group Meeting 1 (Correct data presenting)
Friday 24 th	10:30-12:10		Group Meeting 2 (Write Report)
Monday 27 th	10:30-12:10	QMES Floor 1	Introduction to EXP2-1 Part2
Tuesday 28 th	14:00-15:40		Group Meeting 3 (Different sample comparison)
Friday 3 rd	10:30-12:10		Group Meeting 4 (Complete & submit Report)
Sunday 5 th	17:00	QMPlus	Complete peer assessment

P Groups

Date:	Time:	Place:	Activity:
Monday 20 th	14:00-14:45	QMES Floor 1	Introduction to QXU5017 & EXP2-1 Part1
Tuesday 21 st	10:30-12:10		Group Meeting 1 (Correct data presenting)
Friday 24 th	14:00-15:40		Group Meeting 2 (Write Report)
Monday 27 th	14:00-15:40	QMES Floor 1	Introduction to EXP2-1 Part2
Tuesday 28 th	10:30-12:10		Group Meeting 3 (Different sample comparison)
Friday 3 rd	14:00-15:40		Group Meeting 4 (Complete Report)
Sunday 5 th	17:00	QMPlus	Report submission